

PROJECT: ANALYSING FITBIT DATA

Introduction

In this project, we study a dataset with information obtained from fitbits generated by 33 respondents to an amazon survey in 2016 that submitted censored usage data. The end product will be a dashboard that can be used to study various statistical analysis of the users.

Before starting, please have a careful look at the rules for the [use of ChatGPT and other tools for generative AI](#) and note that you are required to keep a log of your usage of GenAI.

Part 1: Getting acquainted with the data

In the first part, we will just work with the file `daily_activity.csv` that consists of various statistics that the participants of the survey reported daily. In the first part of the assignment, you will explore this dataset and make some basic computations and visualizations.

Basic inspection of the data

Count how many unique users there are in the dataset and then compute the total distance the fitbits registered for each user. Display the results in a suitable graph.

To make it more user-specific, implement a function that takes as input a user Id and displays a line graph that shows the calories burnt on each day. Improve your function to select a specific range of dates in which results are shown. You may want to use the library `datetime` for this purpose.

Use `datetime` or manually look up the day of the week for each date and make a barplot with the day on the x -axis and the frequency at which all individuals work out on the respective day on the y -axis.

Relationship between calories and steps taken

Based on the amount of steps taken, we would like to estimate the calories burnt based on the available data. Next we investigate the relationship between the amount of steps taken and the amount of calories burnt. Use the function `OLS` from `statsmodels` to set up a the following linear regression model

$$\text{Calories} = \beta_0 + \beta_1 \cdot \text{TotalSteps} + \beta_2 \cdot \text{Id} + \epsilon$$

Where $\epsilon \sim N(0, \sigma^2)$. Here, `Id` is taken as explanatory variable as the relationship could possible be different for each individual. Make sure that `Id` is interpreted as a factor here.

Next write a function that take `Id` as input and produces a figure in which the linear relationship for that `Id` is visualized in a scatterplot + regression line.

Use your own creativity to discover any other features about this dataset you find interesting or noteworthy! Look for relations between variables of create attractive visualizations of the data. Points will be awarded for creative insights or features of the final product!

Create one `.py`-file which, upon execution produces all required figures